

Total Lunar Eclipse to Turn Moon Red

A total lunar eclipse is about to make the moon appear a reddish color across the Western Hemisphere.

The event will happen Thursday night into Friday morning. The best places to see the eclipse will be in North America and South America. Parts of Africa and Europe may also get brief views.

A lunar eclipse happens when the sun, Earth, and moon line up just right, with Earth positioned between the sun and moon. This causes the Earth to create a shadow on the moon. In a total lunar eclipse, the Earth's shadow covers all of the moon.

Another kind of eclipse is a solar eclipse. In a solar eclipse, the moon gets in a position where it blocks light from the sun, causing a partial or full shadow on Earth.

A total lunar eclipse can also be called a blood moon. It makes the moon appear reddish-orange, similar to the element copper. The color comes from small amounts of sunlight passing through the Earth's atmosphere.

The American space agency NASA says lunar and solar eclipses happen between four and seven times a year. The last total lunar eclipse was in 2022.

This one will be visible for about one hour starting Friday morning at 2:26 a.m. Eastern Daylight Time (EDT). The time when the Earth's shadow covers all of the moon will be close to 3 a.m. EDT.

"As long as the sky is clear, you should be able to see it," Shannon Schmoll told the Associated Press. She is the director of Abrams Planetarium at Michigan State University. No special equipment will be needed to see the reddish moon.

The total lunar eclipse may be harder to see in Europe and Africa because the moon will be close to setting.

Michael Faison is an astronomy expert from Yale University. He told the AP, "This is really an eclipse for North and South America."

Zoe Ortiz is a historian with the University of North Texas. She noted that different civilizations have observed lunar eclipses for thousands of years. This helped ancient people learn things about the behaviors of the sun, moon, and stars.

“They were looking at the night sky and they had a much brighter vision than we do today,” Ortiz said.

The ancient Greek thinker and writer Aristotle observed that Earth’s shadow on the moon during a lunar eclipse was always curved. This fact supported proof that the Earth is round.

The next total lunar eclipse will appear in the sky September 7, across parts of Asia, Africa, Australia and Europe. Parts of the Americas will get the next chance to see one in March 2026.

I’m Caty Weaver.

Adithi Ramakrishnan wrote this story for The Associated Press. Andrew Smith adapted it for VOA Learning English.

Vocabulary

shadow - n. พื้นที่มืดที่เกิดจากการที่บางสิ่งบางอย่างบดบังแสง

curve - n. เส้นที่โค้งคล้ายกับวงกลม

Reading Comprehension Quiz

1. Which celestial event is primarily discussed in the article?

- A. A solar eclipse
- B. A total lunar eclipse
- C. A partial solar eclipse
- D. A planetary alignment

2. When is the described total lunar eclipse expected to take place?

- A. Friday afternoon
- B. Thursday morning
- C. Thursday night into Friday morning
- D. Saturday evening

3. Which regions are identified as the best places to view this particular lunar eclipse?

- A. Asia and Australia
- B. North America and South America
- C. Europe and Africa
- D. Eastern Hemisphere

4. According to the article, what is the relative position of celestial bodies during a lunar eclipse?

- A. The moon is between the sun and Earth.
- B. The Earth is between the sun and moon.
- C. The sun is between the Earth and moon.
- D. The moon is behind the Earth.

5. How does a solar eclipse differ from a lunar eclipse, as described in the text?

- A. A solar eclipse involves the Earth casting a shadow on the moon.
- B. A solar eclipse occurs when the sun blocks light from the moon.
- C. A solar eclipse happens when the moon blocks sunlight from reaching Earth.
- D. A solar eclipse always results in a full shadow on Earth.

6. What is another common name for a total lunar eclipse, as mentioned in the article?

- A. A copper moon
- B. A red giant
- C. A blood moon
- D. A shadow moon

7. What causes the moon to appear reddish-orange during a total lunar eclipse?

- A. Dust in the moon's atmosphere.
- B. Reflection of light from Earth's oceans.
- C. Small amounts of sunlight passing through Earth's atmosphere.
- D. The moon's own reddish surface minerals.

8. How frequently do lunar and solar eclipses generally occur each year, according to NASA?

- A. Once or twice
- B. Two to three times
- C. Four to seven times
- D. Eight to ten times

9. For how long will the upcoming total lunar eclipse be visible?

- A. Approximately 30 minutes
- B. About one hour
- C. Several hours
- D. A full night

10. What condition is necessary to see the reddish moon, according to Shannon Schmoll?

- A. Using a telescope
- B. A clear sky
- C. Being in a high altitude
- D. Special filtering glasses

11. What equipment is recommended to view this specific lunar eclipse?

- A. A telescope is highly recommended.
- B. Binoculars will enhance the view.
- C. No special equipment is needed.
- D. Specialized filters are essential.

12. Why might the total lunar eclipse be more challenging to observe in Europe and Africa?

- A. The sky will likely be cloudy.
- B. The moon will be too high in the sky.
- C. The moon will be close to setting.
- D. These regions are in the wrong time zone.

13. What is Michael Faison's area of expertise?

- A. History
- B. Astronomy
- C. Meteorology
- D. Geology

14. According to Zoe Ortiz, how did ancient civilizations benefit from observing lunar eclipses?

- A. It helped them predict future natural disasters.
- B. It taught them about the movements of planets.
- C. It helped them learn about the behaviors of the sun, moon, and stars.
- D. It allowed them to develop advanced navigation techniques.

15. How did Aristotle use observations of lunar eclipses to support a scientific theory?

- A. He noted the moon's reddish color indicated atmospheric properties.
- B. He observed the curved shadow of Earth on the moon, supporting the idea of a round Earth.
- C. He calculated the exact timing of eclipses, proving celestial mechanics.

D. He used the eclipse to measure the distance to the moon.